

A COMBINED Z CROSS SECTION FROM CMS OPEN DATA

$Z \rightarrow \mu^+ \mu^-$, $Z \rightarrow \tau_h \tau_h$ and $Z \rightarrow e^+ e^-$ — and what happens when you put them together

16 September 2026 • CMS Open Data 2016 G+H, $\sqrt{s} = 13$ TeV, $L = 16.4 \text{ fb}^{-1}$

The result

- $\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV}) = 1980 \pm 32 \text{ pb}$ per lepton flavour
- the three channels do not agree: $\chi^2/\text{ndf} = 9.67/2$, $p = 0.008$
- $Z \rightarrow ee$ sits 3.0σ above $Z \rightarrow \mu\mu$ — a **fixable convention violation** in the ee fit, not physics (slide 4)
- adding a second *precise* channel finally buys something: $35 \rightarrow 32 \text{ pb}$, 11 % — but the luminosity floor stands

This deck

1. the result, and the problem with it
 2. strategy: three channels, one shared contract
 3. $Z \rightarrow \mu\mu$, $Z \rightarrow \tau_h \tau_h$, $Z \rightarrow ee$
 4. how the combination is actually done
- + backup: references, full breakdown, variations

The measurement at a glance

Data. CMS 2016 Open Data, Run2016G+H, NanoAODv9.
 $\sqrt{s} = 13 \text{ TeV}$, $L = 16\,393.381 \text{ pb}^{-1} \pm 1.2\%$ (normtag).

Three independent measurements, same luminosity, same simulation, same fitter (TRExFitter v1.8.0), same nuisance-parameter names:

- $Z \rightarrow \mu^+ \mu^-$: 10.4 M events, $\delta\sigma/\sigma = 1.8\%$
- $Z \rightarrow e^+ e^-$: 6.3 M events, $\delta\sigma/\sigma = 1.9\%$
- $Z \rightarrow \tau_h \tau_h$: 21 k events, 80% of them jet $\rightarrow\tau_h$ fakes, $\delta\sigma/\sigma = 12\%$

Combination. Covariance (BLUE) of the three, with the shared systematics correlated through the Category contract of fitting/CONVENTIONS.md.

The punchline: the two *precise* channels **disagree by 3.0σ** , and we can say exactly why. Everything else on this slide is downstream of that.

Result

$$\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV}) = 1980 \pm 32 \text{ pb}$$

$$= 1980 \pm 0.5 \text{ (stat)} \pm 22 \text{ (syst)} \pm 7 \text{ (acc)} \pm 21 \text{ (lumi)} \text{ pb}$$

per lepton flavour, assuming lepton universality

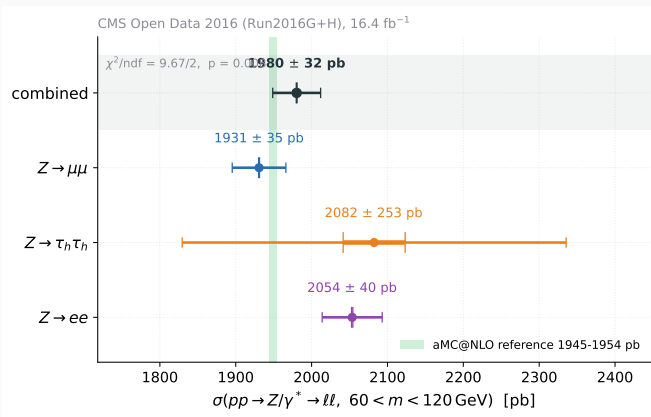
$$\text{channel compatibility } \chi^2/\text{ndf} = 9.67/2, p = 0.008$$

Read this with it

The $Z \rightarrow ee$ fit lets a $\pm 5.9\%$ theory *normalisation* of its own signal template float against its signal strength — forbidden by CONVENTIONS.md §3 precisely because the two are degenerate. Correcting only that moves the combination by -99 pb . **Slide 4.**

Without ee: $1931 \pm 35 \text{ pb}$, $p = 0.54$. CMS measures $1952 \pm 4 \pm 18 \pm 45 \text{ pb}$ (SMP-20-004).

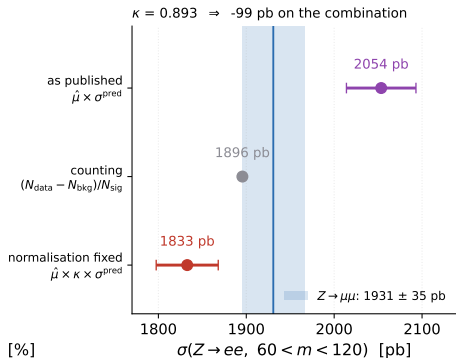
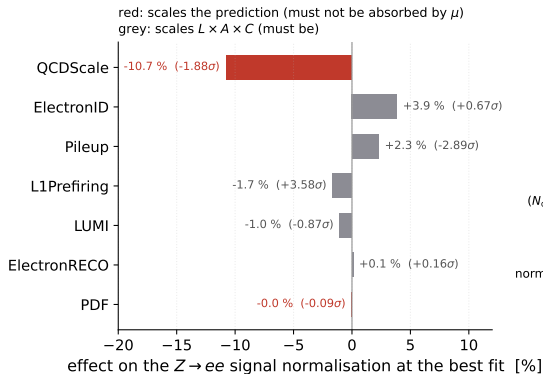
The combined cross section



Weights: $\mu\mu +0.596$, $ee +0.404$, $\tau\tau -0.0003$. The $\tau\tau$ weight is negative — normal BLUE behaviour once ρ exceeds the ratio of the uncertainties — and it moves the central value by 0.06 pb.

$\chi^2/\text{ndf} = 9.67/2$, $p = 0.008$. With two channels this was 1.70/1 ($p = 0.19$) and could be waved through; with three it is a real test and it *fails*. ee and $\mu\mu$ are 123 pb apart at 1.9 % and 1.8 % precision. No PDG scale factor has been applied to hide it.

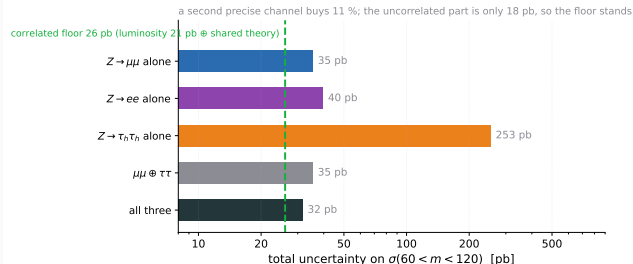
Why $Z \rightarrow ee$ and $Z \rightarrow \mu\mu$ disagree — and it is not physics



CONVENTIONS.md §3: the signal theory templates must **vary the C factor only, renormalised to a constant yield**. $\mu\mu$ and $\tau\tau$ do; ee builds PDF/QCDScale as raw LHE weight envelopes, so QCDScale carries the $\pm 5.87\%$ uncertainty of the aMC@NLO *total cross section* as a normalisation on the signal — exactly degenerate with the POI.

The fit pulls it to -1.88σ : it rescales the *prediction* by $\kappa = 0.893$, and $\text{mu_signal} = 1.0509$ is measured against *that*. The yields say $(N_{\text{data}} - N_{\text{bkg}})/N_{\text{sig}} = 0.9700$, and $\hat{\mu} \times (\text{response of every NP}) = 0.9706$ — closure to 0.05 %. **The fix is one line in the ee notebook**: divide each LHE replica by its own Σw .

Why combining now buys something — and where the floor still is



$Z \rightarrow \mu\mu$ alone: 1931 ± 35.4 pb.

$\mu\mu \oplus \tau\tau$: 1931 ± 35.4 pb — 0.00 %.

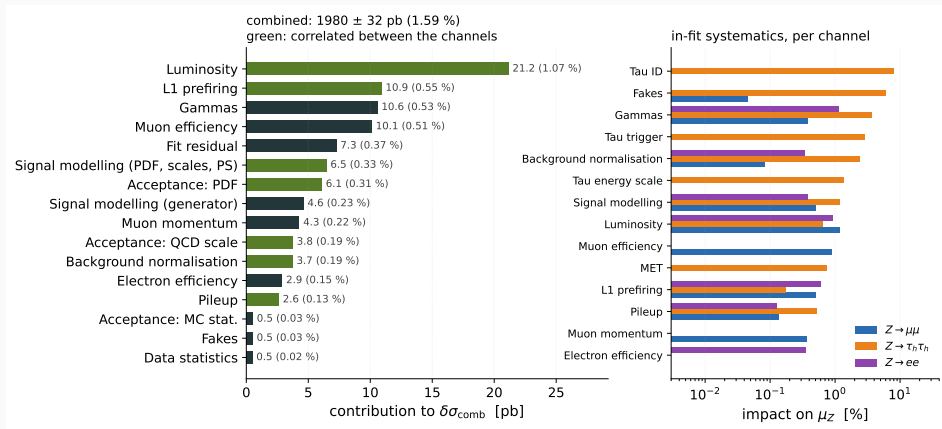
All three: 1980 ± 31.6 pb — 10.7 %.

What changed. $\tau\tau$ is nine times less precise, so it could never move the uncertainty. ee is the first channel of *comparable* precision, and that is exactly what the previous round said was needed.

Why only 10.7 % and not the 29 % two independent equal measurements would give: $\rho(\mu\mu, ee) = 0.44$, almost all of it luminosity.

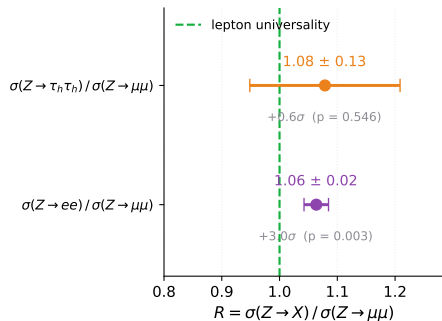
21 pb of the 32 pb is the luminosity: identical normtag value, identical runs, identical $\pm 1.2\%$ in all three channels. A fully correlated uncertainty cannot be combined away. **The route to a better Z cross section from this dataset is a better luminosity calibration, not more channels.**

Where the uncertainty comes from



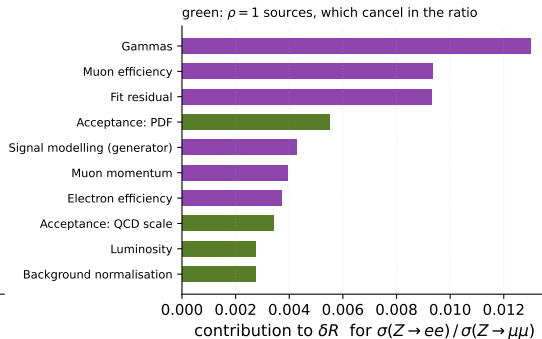
Left: contribution of each source to the combined uncertainty ($\delta_s = \sqrt{w^T C_s w}$); green = correlated between the channels. Right: the same systematics as each channel's fit reports them. Fit residual (7.3 pb) is ee-only: the part of that channel's MINOS error its own grouped impacts do not add up to — carried uncorrelated, so its published total is reproduced exactly.

Lepton universality — and what it is really telling us



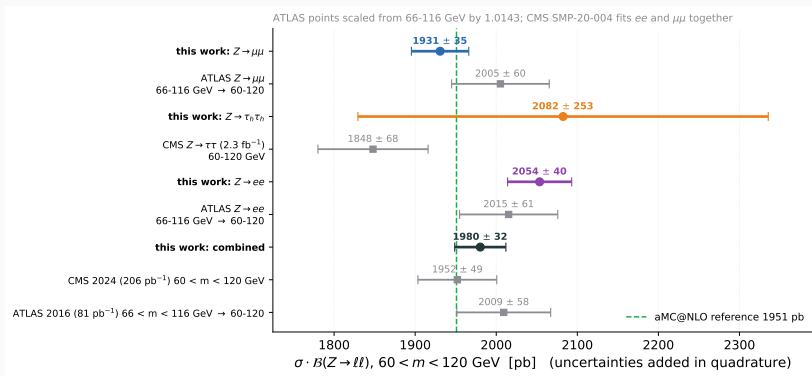
The combination *assumes* universality, so it cannot test it. The ratios can — and in a ratio every $\rho = 1$ source cancels exactly: luminosity, pileup, L1 prefire and the shared PDF/scale/shower terms.

$R(\tau\tau/\mu\mu) = 1.08 \pm 0.13, +0.6\sigma$. It was 1.21 ± 0.17 before $\tau\tau$ moved to DeepTau Tight in v3 — exactly the shift that channel predicted.



$R(ee/\mu\mu) = 1.064 \pm 0.021, +3.0\sigma$. Because so much cancels, the ratio is the *sharpest* statement that the two fits disagree — which is why we quote it as a consistency check between channels and **not** as a limit on $e-\mu$ universality. With the ee normalisation corrected it becomes $0.949 \pm 0.019, -2.7\sigma$: the disagreement changes sign, so “just apply the correction” is not the answer either. A clean ee re-run is.

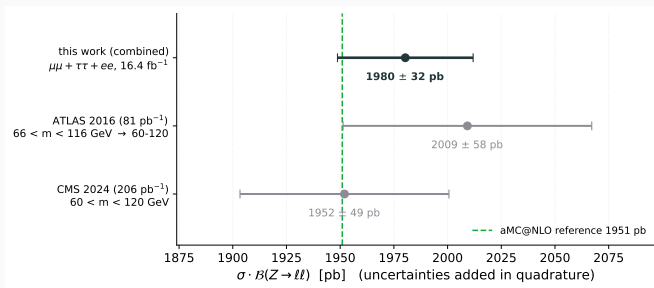
Each channel against the published measurement of the same decay



Read the first two rows together. Each of our two *precise* channels agrees with the ATLAS measurement of the same decay at the 1σ level — $\mu\mu -1.1\sigma$, $ee +0.5\sigma$ — and yet they disagree with *each other* by 3.0σ . No contradiction: ATLAS's per-channel uncertainty is $\pm 60 \text{ pb}$, ours are ± 35 and $\pm 40 \text{ pb}$.

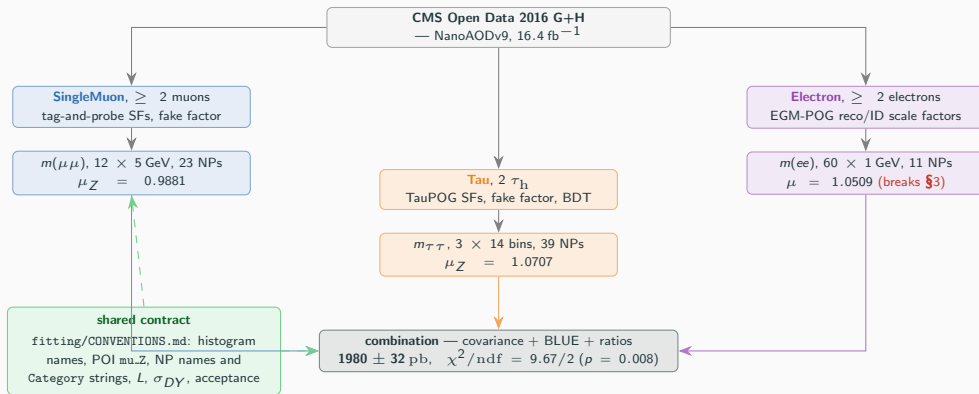
Agreement with a published number is not a validation at this precision. Everything is on our window: ATLAS quotes 66–116 GeV and is scaled by $\sigma(60\text{--}120)/\sigma(66\text{--}116) = 1.0143$ from the same aMC@NLO sample. CMS SMP-20-004 fits ee and $\mu\mu$ *together* and publishes no per-channel number, so ATLAS is the only 13 TeV measurement that can carry this plot. $\tau\tau$: CMS 1801.03535, five final states, same window.

Comparison with published measurements



Agreement with the CMS measurement of exactly the same quantity is within 0.5σ — but so is the $\mu\mu \oplus \tau\tau$ value (1931 pb, 0.4σ) and nearly so the normalisation-corrected one (1882 pb, 1.2σ). **At this precision agreement with a published number does not discriminate between our three readings and must not be used to pick one.** Both published numbers carry a *larger* relative luminosity uncertainty than we do (2.3% and 2.1% against 1.2%) — they are early-Run-2 measurements, while the 2016 G+H normtag luminosity is well measured. Note this is not a fully independent check: the same NNLO reference ($\sigma_{DY} = 6077.22 \text{ pb}$) normalises the simulation behind all three of our C factors.

How the four pieces fit together



The **green** contract is what makes the combination possible at all: because the channels name 'Lumi', 'Pileup', 'L1Prefiring', 'PDF' ... identically and group them into the same Category, the correlations can be assigned category by category instead of guessed. It is also what tells us the ee channel has a problem: §3 of the same contract is the line its theory templates break.

Collision data (Run2016G+H only — 16 of the 36 fb^{-1} CMS recorded in 2016)

dataset	recid	events
SingleMuon G / H	30530 / 30563	324 M
Electron G / H	30529 / 30562	232 M
Tau G / H	30532 / 30565	156 M

Certified with the 2016 golden JSON restricted to runs 278820–284044. The three *selections* are disjoint: the only process that can enter two of them is $ZZ \rightarrow 4\ell$, 61 of the 10.4 M $\mu\mu$ events (0.0006 %), so the data statistics are uncorrelated as a fact (backup).

Luminosity. $L = 16\,393.381\text{ pb}^{-1}$, the *normtag* value from CMS Open Data record 1059. An earlier *brilcalc* call without `--normtag` gave $16\,290.713\text{ pb}^{-1}$, 0.63 % low; all three channels use the *normtag* value, which is why the luminosity is *exactly* correlated between them — and at 1.2 % it is better than the 2.3 % CMS and 2.1 % ATLAS carry.

Simulation (UL16 postVFP, streamed from EOS or read from dCache)

sample	recid	used for
DY M-50 aMC@NLO	35669	signal, all three
DY M-50 madgraph	35671	generator syst.
DY M-10to50	35631	$\tau\tau$ background
$t\bar{t}$, tW , VV , W +jets	—	backgrounds

Normalisation. $\sigma(Z/\gamma^* \rightarrow \ell\ell, m > 50\text{ GeV}) = 6077.22\text{ pb}$ (NNLO). Each channel's μ_Z multiplies the LHE-flavour slice of it inside $60 < m_{\text{LHE}} < 120$:

$\sigma^{\text{pred}}(60\text{--}120)$	
ee	1954.1 pb
$\mu\mu$	1953.9 pb
$\tau\tau$	1944.9 pb

The 0.47 % spread is the generator's flavour shares (the τ mass is in the matrix element), not a definition — which is why we combine *cross sections*, never μ_Z (backup).

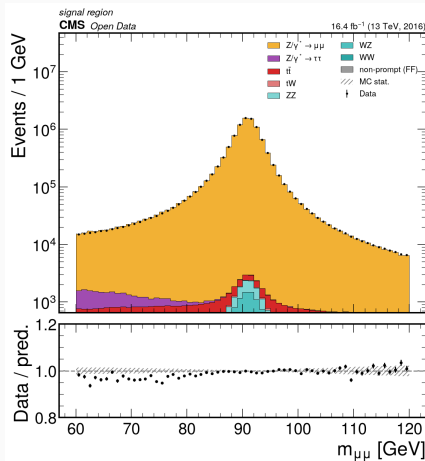
$Z \rightarrow \mu^+ \mu^-$: a 1.8 % measurement with 10 million events

Selection. HLT_IsoMu24 || HLT_IsoTkMu24; exactly two tight, isolated, opposite-sign muons, $p_T > 26/20$ GeV, $|\eta| < 2.4$, $60 < m(\mu\mu) < 120$ GeV; FSR photons recovered from the FsrPhoton collection.

Corrections, all measured in this data.

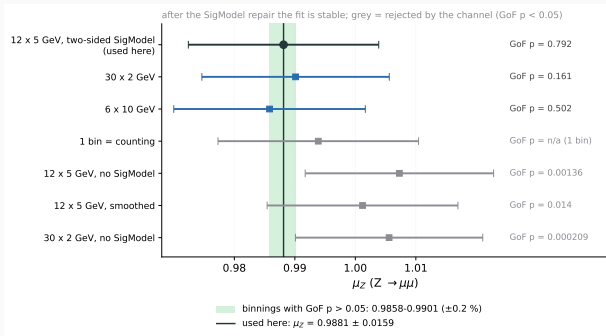
- ID / isolation / trigger scale factors from tag-and-probe fits on the Z peak (20 M pairs), with trigger-object matching
- pileup weights from the Open Data luminosity table, L1-prefiring weights
- Z-peak muon momentum calibration per $|\eta|$ bin
- non-prompt muons from a same-sign fake factor (3.8 k events, 0.04 % of the yield)

Backgrounds are tiny: 69 k of 10.4 M events (0.66 %), dominated by $t\bar{t}$ (32 k) and $Z \rightarrow \tau\tau$ (11 k).



The distribution that is fitted, shown in 1 GeV input bins; the fit runs in 12×5 GeV. Note the ratio panel — data sit 3–4 % below the prediction at 62–78 GeV. That deficit is the reason the next slide exists.

$Z \rightarrow \mu\mu$: which number to take — the channel's review, and its repair



The channel's own review (`z-mumu/REVIEW.md`, F3/F4) found the 30-bin shape fit **not robust**: μ_Z moved over 0.990–1.006, because the only template that could absorb a 3–4 % data deficit at 62–78 GeV was SigModel — and that template mixed the generator comparison with a phase-space artefact (the powheg sample is generated with $m_{\text{LHE}} < 120$ GeV, depleting its two highest bins by 10 % and 33 %).

The channel fixed both. Fit in 12×5 GeV bins so the shape parameters are not over-constrained, no smoothing, MINOS everywhere; SigModel rebuilt with both generators inside a common $50 < m_{\text{LHE}} < 120$ GeV window and mirrored.

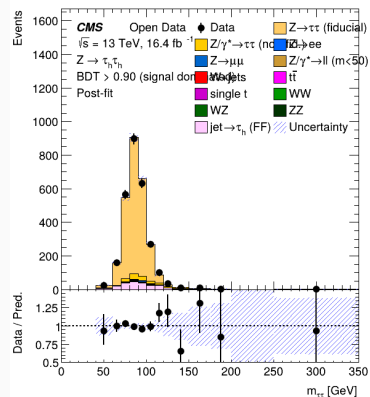
We therefore take the shape fit: $\mu_Z = 0.9881 \pm 0.0159$, GoF $p = 0.79$, stable to $\pm 0.2\%$ across the binnings that describe the data. The $\pm 0.7\%$ lineshape term the review had asked for as a stop-gap is **retired** — the two-sided template carries it inside the fit. Counting instead would move the combination by +11.2 pb: reported as a variation, not used.

$Z \rightarrow \tau_h \tau_h$: a 12 % measurement where 80 % of the data is background

Selection. Double- τ_h trigger (different path in G and H), both legs HLT-matched; two opposite-sign τ_h with $p_T > 40$ GeV, $|\eta| < 2.1$, decay modes 0/1/10/11, DeepTau2017v2p1 **Tight** vs jets (v3 — the working point v2.1 recommended and v3 adopted); e/μ veto keeps it orthogonal to the other two channels. $\Rightarrow$ 21 160 events, of which $\sim 64\%$ are jets faking a τ_h .

No QCD simulation is used. The fake background comes from data: a fake factor $FF(era, DM, N_{jets}, p_T)$ measured in a same-sign region and applied to opposite-sign events whose leading τ_h fails Tight, times an OS/SS correction of 1.04–1.15 measured per $(era, N_{jets}, BDT \text{ category})$, with the genuine- τ MC subtracted. A k -fold BDT (held-out AUC 0.966) splits the signal region into **three categories** by purity.

Mass. The visible di- τ mass is not a Z mass. A per-event likelihood over the τ visible-energy fractions using the MET covariance gives $m_{\tau\tau}$ with scale 0.99 and resolution 11 % (against 0.80 / 13 % for m_{vis}).



Post-fit $m_{\tau\tau}$ in the most signal-like BDT category (score > 0.90); the fit runs over all three, 14 bins each, 0–350 GeV. Pink = the data-driven jet $\rightarrow \tau_h$ fakes, orange = $Z \rightarrow \tau\tau$. Counting over the whole signal region instead gives 2173 pb: it is the low-score categories that fix the fake normalisation.

$Z \rightarrow \tau_h \tau_h$: result and what limits it

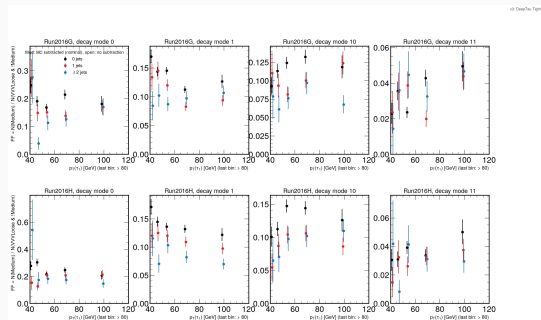
$$\mu_Z = 1.071^{+0.114}_{-0.100}$$

$$\sigma(60-120) = 2082 \pm 41^{+222}_{-194} \pm 76 \text{ pb}$$

goodness of fit $p = 0.25$

source	impact on μ_Z
τ_h ID	8.4 %
fakes (fake factor)	6.5 %
MC statistics	3.9 %
τ_h trigger	3.0 %
background normalisation	2.6 %
τ_h energy scale	1.5 %
signal modelling (PDF, scales, PS)	1.3 %
luminosity	0.7 %
data statistics	2.1 %

Acceptance $A = 0.23\%$ with a 3.7% uncertainty — the double 40 GeV cut on the *visible* τ momentum sits on the Z p_T tail, where the QCD scale dependence is 3.4% .



The fake factors, binned in era, decay mode, N_{jets} and p_T . Without the N_{jets} binning the same-sign closure is off by 10–25 % per jet bin; without the era binning by $\pm 5\%$ (the HLT τ isolation changed between G and H).

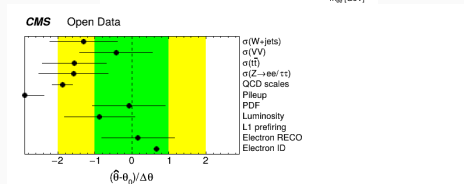
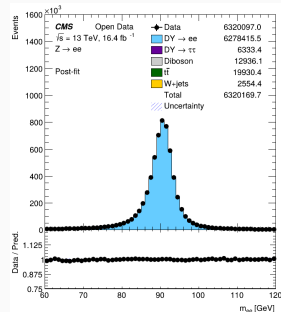
Moving from DeepTau Medium (v2.1) to Tight (v3) took the channel from $\mu_Z = 1.205$, $\delta\sigma/\sigma = 14\%$ to $\mu_Z = 1.071$, 12% — and $R(\tau\tau/\mu\mu)$ from 1.21 ± 0.17 to 1.08 ± 0.13 .

$Z \rightarrow e^+e^-$: 6.3 million events, and one line of the contract

Selection. HLT_Ele27_WPTight_Gsf; exactly two opposite-sign electrons, $p_T > 20$ GeV, $|\eta| < 2.5$, cutBased $\geq$ Medium; $60 < m_{ee} < 120$ GeV. $\Rightarrow$ **6 320 097** events, background 44.9 k (0.7%): $t\bar{t}$ 21.6 k, diboson 13.2 k, $Z \rightarrow \tau\tau$ 6.7 k, W +jets 3.4 k.

Corrections. Pileup, L1-prefiring, and EGM-POG reco + Medium-ID scale factors. DY is split by LHE truth, so the $\tau\tau$ feed-down is a separate template. **No trigger SF, no electron energy scale, no FSR recovery, no charge-misID, no fake background** (backup).

Fit. TRExFitter v1.8.0, m_{ee} in 60×1 GeV bins, one signal region, 11 NPs + 60 γ . $\Rightarrow \mu = 1.0509 \pm 0.0202$, $\sigma(60-120) = 2054 \pm 40$ pb. **No goodness of fit was run** — the only channel without one.



Pileup at -2.9σ , L1 prefiring at $+3.6\sigma$ (off the axis), Electron ID squeezed to 0.08 of its input width: a shape mismatch absorbed by calibration parameters in a 1 GeV binning. $z\text{-}\mu\mu$ fixed the same thing with 5 GeV bins.

What is combined, and how

The observable.

$\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV})$ per lepton flavour, assuming lepton universality:

$$\sigma_c = \hat{\mu}_c \times \sigma_c^{\text{pred}}(60-120)$$

Not μ_Z itself: the three channels normalise to references 9 pb apart for the same quantity (backup), so averaging the μ 's would average three different things.

The covariance, source by source,

$$C = \sum_s C_s, \quad (C_s)_{ij} = \rho_s s_i s_j$$

then standard BLUE,

$$w = \frac{C^{-1}u}{u^T C^{-1}u}, \quad \hat{\sigma} = w^T x, \quad \text{Var} = \frac{1}{u^T C^{-1}u}$$

with χ^2 (ndf = 2 now) for compatibility, and the combined uncertainty decomposed as $\delta_s = \sqrt{w^T C_s w}$.

Two details that are easy to get wrong.

In-fit systematics scale with the prediction, because μ_Z multiplies a fixed number: $\delta\sigma = s_{\text{group}} \times \sigma^{\text{pred}}$. *The acceptance is multiplicative*, $\delta\sigma = a \times \sigma^{\text{meas}}$, since $\sigma = \sigma_{\text{fid}}/A$ — and A is outside the $\mu\mu$ and $\tau\tau$ likelihoods (0.61 % and 3.7 %), so the combination has to add it. The ee channel fits against the full 60–120 GeV prediction directly, so for it A is *inside* the fit and there is no separate term.

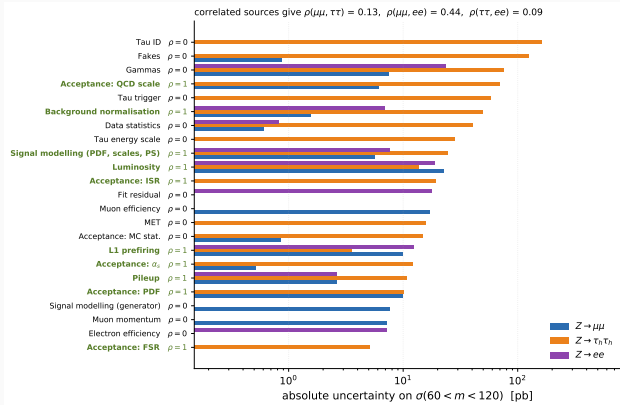
Because the acceptance is multiplicative, the BLUE is *iterated*: the terms are re-evaluated at the combined value until stable (< 0.1 pb).

Cross-checked against an explicit profile likelihood with one nuisance per shared source,

$$-2 \ln L = \sum_c \frac{(x_c - \sigma - \sum_s d_{cs} \theta_s)^2}{u_c^2} + \sum_s \theta_s^2$$

which additionally carries the $\tau\tau$ channel's asymmetric error. Same answer: 1980^{+31}_{-31} against 1980 ± 32 pb.

The correlation model — the only judgement in the whole thing



$\rho = 1$ — derived once, used by all

- Luminosity: same normtag value, same runs, same $\pm 1.2\%$
- Pileup, L1 prefire: same 2016 profile, same weights
- Background normalisation: literally the same XS_* parameters on the same MC samples
- PDF, α_s , QCD scale, PS: same Hessian members and the same 7-point envelope of the *same* aMC@NLO sample

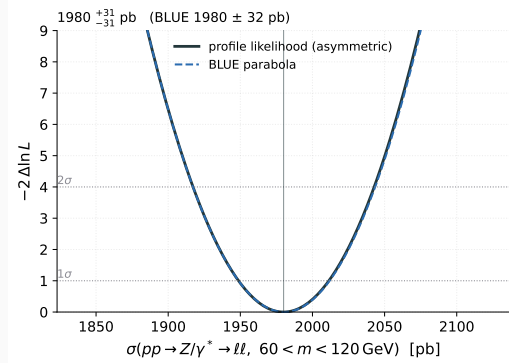
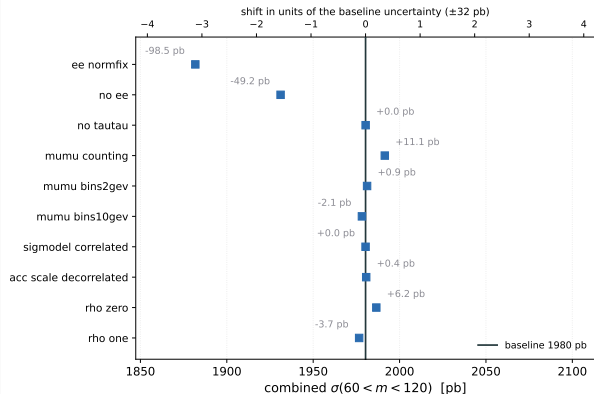
$\rho = 0$ — different objects, methods or events

- muon vs electron vs τ_h calibrations: no common input
- Gammas: same sample, three *disjoint* selections
- fakes, MET; Fit residual (*ee* only)
- data statistics: *disjoint* primary datasets

The split of Signal modelling. Only $\mu\mu$ still fits a generator systematic (powheg vs aMC@NLO, 0.4% on μ_Z); $\tau\tau$ dropped its LO-vs-NLO version in v2.1 and *ee* never had one. We still split the category and decorrelate the generator part — so the “SigModel correlated” variation is exactly null.

$\Rightarrow \rho(\mu\mu, ee) = 0.44$ (almost all luminosity), $\rho(\mu\mu, \tau\tau) = 0.13$, $\rho(\tau\tau, ee) = 0.09$.

Cross-checks: the model is not what limits us — one channel is



The two rows that *bracket the entire correlation model* span 9.9 pb, under a third of the total uncertainty, and every modelling choice inside that bracket moves the answer by less than 0.4σ . What dominates instead is **ee normfix at -98.5 pb (-3.1σ)** — a convention violation inside one channel — and **no ee at -49.2 pb**, the size of the ee contribution. Right: the profile likelihood and the BLUE parabola lie on top of each other.

What this is not, and what is left to do

Not a TRExFitter MultiFit

The channel workspaces are not all committed and were not rebuilt, so the shared nuisance parameters are correlated through their Category totals rather than *profiled jointly*. The fitting is still TRExFitter's — it produced all three inputs.

The MultiFit config is generated and syntax-checked (`combination/fit/comb.config`); running `fit/run_multifit.py --dry-run` now prints what still blocks it:

```
POI: mu_Z           (z-ee: mu_signal)
Region: ee_SR        (z-ee: "SR")
Sample DYee, SIGNAL (z-ee: DY_ee, BACKGROUND)
Lumi, XS_WW/WZ/ZZ   (z-ee: LUMI, XS_VV)
PDF/QCDScale renormalised to a
    constant yield   (z-ee: raw envelopes)
```

This matters more than it used to: ee carries 40 % of the weight, and a joint fit would expose the QCDScale problem *structurally* instead of as a variation.

What $Z \rightarrow ee$ has to change (`combination/docs/01-inputs.md`)

1. **renormalise PDF/QCDScale to a constant yield** — this is the whole 99 pb
2. accumulate the envelopes per LHE flavour
3. adopt `CONVENTIONS.md` §1–§4 and publish a results JSON
4. run the saturated-model goodness of fit
5. fit in coarser bins than 60×1 GeV
6. smooth or merge W jets (24 % MC stat., ~ 30 floored bins)

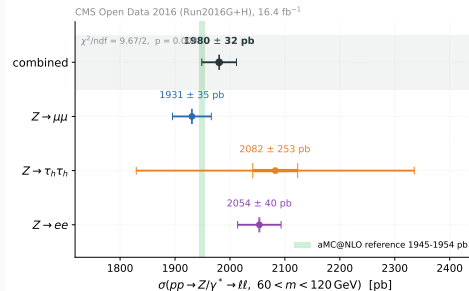
Already fixed on 16 Sep: the reference cross section (their 1954.1032 pb agrees with ours to 4×10^{-6}) and the negative bins. Full list: `ASK_Z_EE.md`.

Inherited caveats we did not fix: the $\mu\mu$ 60–80 GeV lineshape; prefiring, muon and pileup calibrations all in-house; $\tau\tau$ v3 publishes only its nominal, so there are no $\tau\tau$ cross-check rows.

Conclusions

$$\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV}) =$$
$$1980 \pm 32 \text{ pb}$$
$$\chi^2/\text{ndf} = 9.67/2 \quad (p = 0.008)$$

1. **Quote the compatibility with the value.** Three very different analyses of the same 16.4 fb^{-1} , and the two *precise* ones are 123 pb apart.
2. **We know why, and it is fixable.** The ee fit lets a $\pm 5.9\%$ theory *normalisation* float against its own signal strength; QCDScale is pulled -1.88σ and $\hat{\mu}$ is measured against a prediction the fit rescaled by 0.893. Correcting only that: -99 pb . **One line in the ee notebook.**
3. **Combining finally buys something:** $35 \rightarrow 32 \text{ pb}$, 11 % — the first channel of precision comparable to $\mu\mu$ does what $\tau\tau$ never could. But 21 pb of the 32 pb is **luminosity**, fully correlated and irreducible.
4. Data statistics are 0.5 pb; MC statistics are 10.6 pb. **More collisions would change nothing**; more simulated events would.
5. The shared conventions made this possible — and **the one place a channel departed from them is the biggest number in the cross-check table.**



Without ee : $1931 \pm 35 \text{ pb}$, $p = 0.54$. With the ee normalisation corrected: $1882 \pm 30 \text{ pb}$, $p = 0.025$. All numbers and code:

```
combination/result.md, combination/docs/, python  
run_combination.py.
```

Backup

Backup: the three references differ by 0.47 %

All three channels define μ_Z as a multiplier on an aMC@NLO prediction of $\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV})$, and all three now build it the same way — $6077.22 \text{ pb} \times \Sigma w(\text{LHE flavour}, 60 < m_{\text{LHE}} < 120)/\Sigma w$ of the same sample. The generator's flavour shares are simply not exactly 1/3:

channel	$\Sigma w_{\text{flav}}/\Sigma w$	$\sigma^{\text{pred}}(60\text{--}120)$
ee	0.33386	1954.1 pb
$\mu\mu$	0.33380	1953.9 pb
$\tau\tau$	0.33234	1944.9 pb

ee and $\mu\mu$ agree to 0.01 %; $\tau\tau$ is 0.44 % lower because the τ mass is in the matrix element. With 72 M generated events the statistical precision on any of them is 0.02 %, so the spread is **physical**, not definitional.

Why it matters.

- We combine *cross sections*: each $\hat{\mu}$ is multiplied by its own reference first, so the spread shows up as a band on the theory line rather than as a bias.
- A TRExFitter MultiFit with one shared `mu_Z` would *not* handle it — it would fit one parameter against three references.
- Size today: $\sim 0.002\%$ on the combination. Worth fixing before a joint fit is quoted, not before this one.

$\mu\mu$ reaches its number by the equivalent route $\sigma_{\text{fid}}^{\text{pred}}/A(60\text{--}120) = 799.566/0.409209$; z-ee computes its own since 16 Sep 2026 and agrees with our independent value to 4×10^{-6} . Recorded as a decision

in `fitting/CONVENTIONS.md` §6.

Backup: full uncertainty breakdown of the combined value

source	pb	of σ	corr.
Luminosity	21.15	1.07 %	yes
L1 prefiring	10.92	0.55 %	yes
Gammas (MC stat.)	10.59	0.53 %	no
Muon efficiency	10.13	0.51 %	no
Fit residual (ee)	7.27	0.37 %	no
Sig. modelling (PDF, scales, PS)	6.47	0.33 %	yes
Acceptance: PDF	6.10	0.31 %	yes
Sig. modelling (generator)	4.61	0.23 %	no
Muon momentum	4.27	0.22 %	no
Acceptance: QCD scale	3.75	0.19 %	yes
Background normalisation	3.74	0.19 %	yes
Electron efficiency	2.89	0.15 %	no
Pileup	2.62	0.13 %	yes
Acceptance: MC stat.	0.53	0.03 %	no
Fakes	0.52	0.03 %	no
Data statistics	0.49	0.02 %	no
Acceptance: α_s	0.31	0.02 %	yes
Tau ID	0.05	0.00 %	no
total	31.6	1.59 %	

Contributions add in quadrature: $\delta_s = \sqrt{w^T C_s w}$ with C_s the covariance of that source alone.

Reading the table:

- the top entry and four of the top ten are **correlated**, so they set a floor combining cannot touch — 26 pb of the 31.6 pb;
- Fit residual is ee-only: the part of that channel's published MINOS error its own grouped impacts do not add up to (0.88 % of $\hat{\mu}$). TRExFitter's grouped impacts are quadrature differences and need not close; $\mu\mu$ and $\tau\tau$ over-shoot, ee under-shoots. Carrying it uncorrelated reproduces the published total exactly;
- everything the $\tau\tau$ channel contributes sums to 0.06 pb;
- the data statistical uncertainty is 0.5 pb, four orders of magnitude below the cross section — and *twenty times smaller* than the MC statistics.

Generated table: `combination/result.md`, machine-readable version
`combination/output/combination.result.json`.

Backup: variations in numbers

variation	σ [pb]	shift	what it changes
baseline	1980.3	—	$\mu\mu$ shape fit, $\tau\tau$ v3, ee as published
ee normfix	1881.8	−98.5	ee PDF/QCDScale normalisation put back in the prediction
no ee	1931.1	−49.2	the $\mu\mu \oplus \tau\tau$ combination
no $\tau\tau$	1980.4	+0.0	the two precise channels only
$\mu\mu$ counting	1991.4	+11.1	1-bin extraction instead of the 12×5 GeV fit
$\mu\mu$ 30×2 GeV	1981.2	+0.9	alternative binning (GoF $p = 0.16$)
$\mu\mu$ 6×10 GeV	1978.2	−2.1	alternative binning (GoF $p = 0.50$)
SigModel correlated	1980.3	+0.0	generator comparison shared — null, see §4
acc. scale decorrelated	1980.7	+0.4	acceptance QCD-scale term independent
$\rho = 0$ everywhere	1986.5	+6.2	all systematics uncorrelated (naive)
$\rho = 1$ everywhere	1976.6	−3.7	all systematics fully correlated

The $\rho = 0$ and $\rho = 1$ rows bracket every possible correlation assignment: they span 9.9 pb, 0.31σ . Every modelling choice inside that bracket moves the answer by less than 0.4σ . The two entries that dominate the table are not modelling choices at all: what one channel does to its own signal template (−98.5 pb) and whether that channel is in (−49.2 pb). χ^2/ndf : 9.67/2 baseline, 7.39/2 with normfix, 0.37/1 without ee.

Backup: the channels' own numbers

	$Z \rightarrow \mu^+ \mu^-$	$Z \rightarrow \tau_h \tau_h$	$Z \rightarrow e^+ e^-$
primary dataset	SingleMuon	Tau	Electron
selected events	10 378 567	21 160	6 320 097
background	68 787 (0.66 %)	16 959 (80 %, prefit)	44 912 (0.71 %)
A (fiducial / 60–120)	0.4092	0.002314	in the fit
C (reco / fiducial)	0.7914	0.0510	in the fit
fit variable	$m(\mu\mu)$, 12×5 GeV	$m_{\tau\tau}$, 3×14 bins	$m(ee)$, 60×1 GeV
nuisance parameters	23 (+12 γ)	39 (+42 γ)	11 (+60 γ)
σ^{pred} (60–120)	1953.9 pb	1944.9 pb	1954.1 pb
μ_Z	0.9881 ± 0.0159	$1.071^{+0.114}_{-0.100}$	1.0509 ± 0.0202
σ_{fid}	790.1 pb (pred. 799.6)	4.82 pb (pred. 4.50)	—
σ (60–120)	1931 ± 35 pb	2082 ± 253 pb	2054 ± 40 pb
goodness of fit	$p = 0.79$	$p = 0.25$	not run
BLUE weight	+0.596	−0.0003	+0.404

The asymmetry is the point: two are precision measurements whose difficulty is calibration, the third is a needle in a haystack whose difficulty is the background. They share a luminosity, a signal sample and a fitter, and nothing else — which is why a shared naming contract was worth writing, and why the one place it was not followed costs more than every modelling choice together.

Backup: are the three channels really orthogonal?

The test is the *selection*, not the dataset: in CMS an event that fires two triggers lands in two primary datasets.

	$\mu\mu$	$\tau\tau$	ee
required	exactly 2 tight μ $p_T > 26/20$	exactly 2 Tight τ_h $p_T > 40$	exactly 2 medium e $p_T > 20$
e veto	no	yes	—
μ veto	—	yes	no
τ_h veto	no	—	no

$\tau\tau$ is disjoint from both by construction: it vetoes any medium muon and any WP90 electron above 10 GeV, which every $\mu\mu$ and ee event has two of.

$\mu\mu \cap ee$ is not empty. Neither vetoes the other's flavour, so an event can have two tight muons *and* two medium electrons — which needs four leptons, so only $ZZ \rightarrow 4\ell$. Measured on the ZZ_4L skim: of the 1016 $ZZ \rightarrow 4\ell$ events in the $\mu\mu$ SR, **61** also pass the ee selection.

The overlap, in numbers

61 events
= 0.0006 % of the 10.4 M $\mu\mu$ events
= 0.001 % of the 6.3 M ee events
induced $\rho(\text{stat}) \sim 8 \times 10^{-6}$

So Data statistics gets $\rho = 0$ as a *measured fact*, and `rho_override` deliberately does not touch it.

One caveat worth stating: $\tau\tau$ has 110 ± 13 prefit DYee events in its SR ($Z \rightarrow ee$ with electrons faking τ_h). Those survived the electron veto *because* they were not reconstructed as electrons, so they cannot also satisfy “exactly two medium electrons”. Even counting all 110 gives 0.5 % of the $\tau\tau$ SR and 0.002 % of the ee SR.

What would break this: adding $e\tau_h$ or $\mu\tau_h$. Those share a lepton leg and a primary dataset with $\mu\mu/ee$ and would need an explicit overlap removal.

Backup: systematics CMS and ATLAS have that we do not

The same sources, different names

	CMS SMP-20-004	ATLAS 1603.09222
lepton reco/ID/iso	$0.28 \oplus 0.16$	$0.9 \oplus 0.3$
lepton trigger	in "efficiency"	$0.1 / 0.2$
lepton energy scale	shape NP	$0.2 / 0.1$
charge misID	—	0.1 (ee)
L1 prefiring	0.34	n/a
pileup	in efficiency	< 0.1
MC statistics	0.16	—
EW + $t\bar{t}$ σ	0.04	in bkg
QCD multijet	$0.14 \oplus 0.07$	in bkg
PDF + α_S	0.43	$0.1 \oplus$ in A
μ_R, μ_F	0.66	in A
resummation + FSR	0.12	in A
total syst.	0.90	1.0 / 1.1 ($\oplus$ 1.8 on A)
luminosity	2.3	2.1
ours		1.2

All in %. Our luminosity is the best of the three: the final 2016 normtag.

What we are missing — five, four of them in ee:

1. **electron trigger efficiency**: ee applies no trigger SF and assigns no uncertainty ($\mu\mu$ and $\tau\tau$ both do)
2. **electron energy scale/resolution**: none in ee — the one systematic that moves the shape of the peak being fitted
3. **charge misidentification**: ee requires OS, assigns nothing; ATLAS quotes 0.1 %
4. **FSR**: ee has no photon recovery and no FSR uncertainty — and electrons radiate more than muons
5. **multijet/fake background**: ee has no data-driven estimate; CMS quotes 0.14 %

Together well under 1 %, so they do *not* explain the 6 % $ee-\mu\mu$ gap. Listed because a reviewer will ask.

And the one that should embarrass us. CMS constrains the τ_h ID efficiency to **2.2 %** by fitting five $\tau\tau$ final states together and promoting that nuisance parameter to a POI. We quote **8.4 %**, straight from the TauPOG scale factors. That single difference is most of why CMS reaches 3.1 % systematic on $\sigma(Z \rightarrow \tau\tau)$ and we reach 12 % — and it is exactly what adding $e\tau_h/\mu\tau_h$, or a MultiFit, would buy.

Backup: where everything lives

combination/result.md	the result, generated from the JSON — never edited by hand
combination/docs/00--05	overview, inputs, correlations, method, results, vs. published
combination/comb/	inputs, model, blue, likelihood, ratio, plots, report
combination/run_combination.py	orchestrator; --check runs 8 closure tests
combination/tools/extract_zee.py	z-ee's TRExFitter job → inputs/zee_fit_result.json
combination/fit/comb.config	the TRExFitter MultiFit, generated and validated, not run
combination/output/	combination_result.json, plots/*.pdf

z-mumu/, z-tau/tau/, z-ee/	the channel analyses, each with its own README/handoff.md
z-mumu/REVIEW.md	the review, and the fixes that made the shape fit usable
fitting/CONVENTIONS.md	the shared contract; §6 records what the combination decided
ASK_Z_EE.md	the list of changes z-ee needs, biggest first
combination/presentation/	this deck (bash build.sh)

Closure tests have been checked, each published measurement with re-assigned a single channel BLUE weights; reproduces the combination exactly; each published measurement with re-assigned a single channel BLUE weights; with every systematic correlated only the statistical part averages; the profile likelihood reproduces BLUE to 10^{-3} pb; dropping a channel reproduces the sub-combination exactly; and the ee grouped impacts plus Fit residual plus the analytic Poisson statistics reproduce that channel's published MINOS error.